Violating the assumptions of the CLRM

Econometrics. Lecture 6

The CLRM assumptions. Review

Assumption MLR.1 (Linear in parameters)	$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$
Assumption MLR.2 (Random sampling)	A random sample of *n* observations, $\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i): i = 1, 2, \ldots, n\}$
Assumption MLR.3 (No perfect collinearity)	None of the independent variables is constant, and there are no exact linear relationships among them
Assumption MLR.4 (Zero conditional mean)	$\mathrm{E}(u \mid x_1, x_2, \ldots, x_k) = 0$
Assumption MLR.5 (Homoskedasticity)	$\mathrm{Var}(u \mid x_1, x_2, \ldots, x_k) = \sigma^2$
Assumption MLR.6 (Normality)	$u \sim \mathrm{Normal}(0, \sigma^2)$.

Content

1. Nonrandom samples
2. Multicollinearity
3. Heteroskedasticity
4. Normality of residuals

1. Nonrandom samples

Exogenous sample selection

Suppose that we are estimating a saving function, where annual saving depends on income, age, family size, and perhaps some other factors.

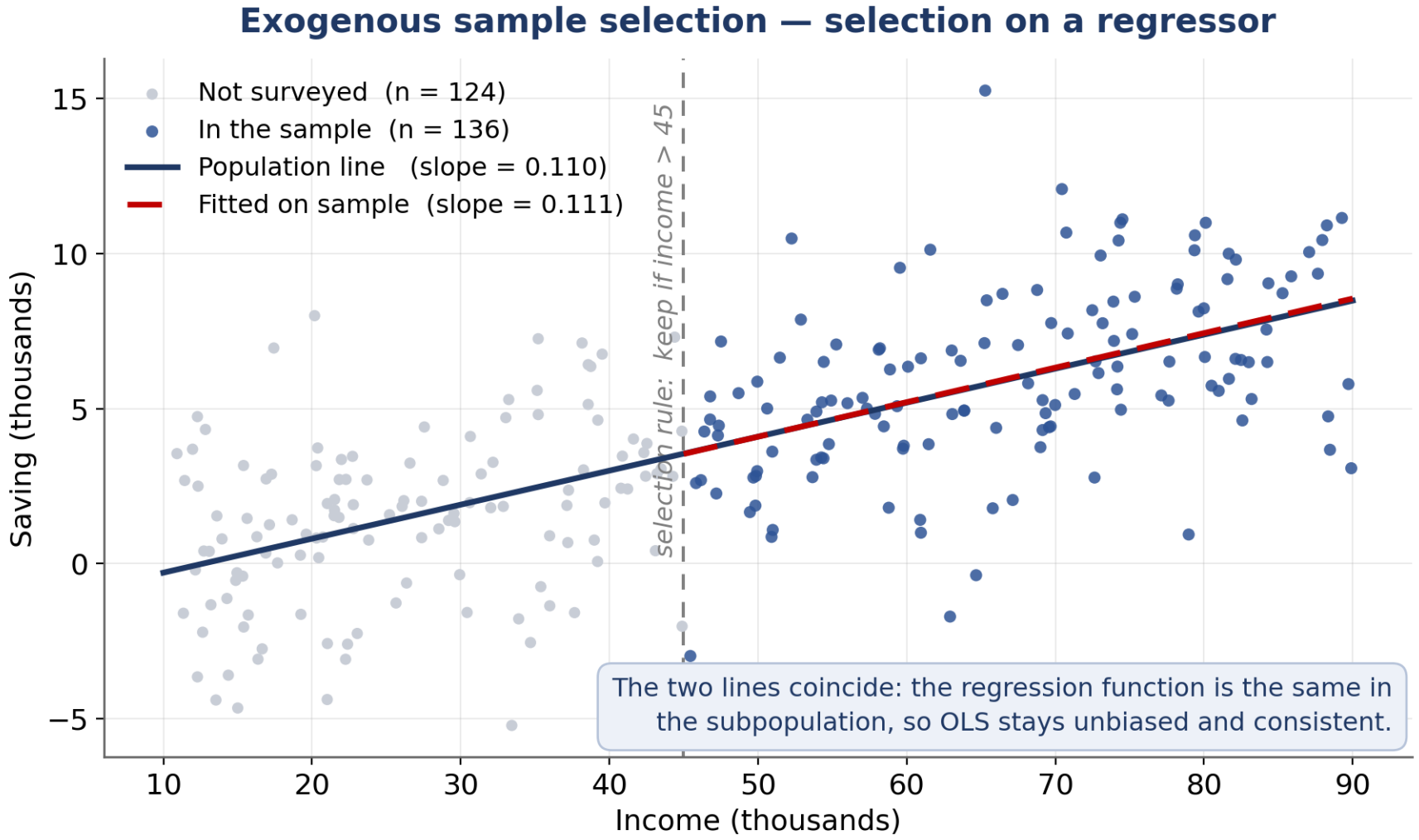
$$saving = \beta_0 + \beta_1 income + \beta_2 age + \beta_3 size + u$$

Suppose that our data set was based on **a survey of people over 35 years of age**, thereby leaving us with a nonrandom sample of all adults.

While this is not ideal, we can still get unbiased and consistent estimators of the parameters in the population model, using the nonrandom sample.

Provided there is enough variation in the independent variables in the subpopulation, **selection on the basis of the independent variables** is not a serious problem, other than that it results in smaller sample sizes.

Exogenous sample selection. An illustration



Endogenous sample selection

If the **sample** is **based on** whether the **dependent variable** is above or below a given value, **bias always occurs** in OLS in estimating the population model.

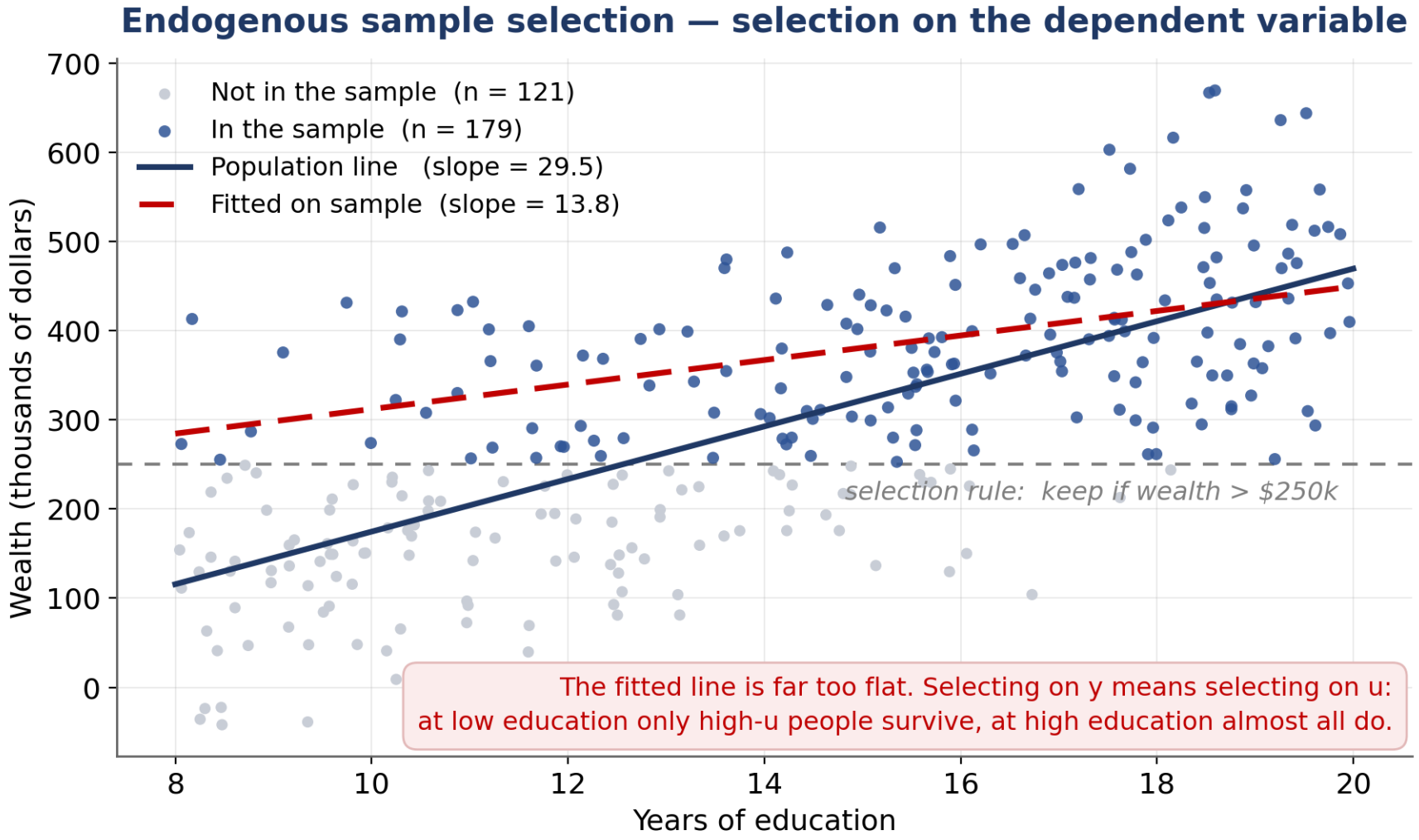
Suppose we wish to estimate the relationship between individual wealth and several other factors in the population of all adults

$$wealth = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 age + u$$

Suppose that only people with wealth over $250,000 are included in the sample.

Using a sample on people with wealth over $250,000 will result in biased and inconsistent estimators of the parameters.

Endogenous sample selection. An illustration



2. Multicollinearity

Multicollinearity

A high degree of linear relationship between independent variables can lead to large variances for the OLS slope estimators. Adding an irrelevant variable to an equation generally increases the variances of the remaining OLS estimators.

High (but not perfect) correlation between two or more independent variables is called **multicollinearity**.

Example of multicollinearity is estimation of consumption with income and wealth being independent variables.

- Income and wealth may be highly correlated.

Consequences of multicollinearity

Multicollinearity violates none of the MLR assumptions!

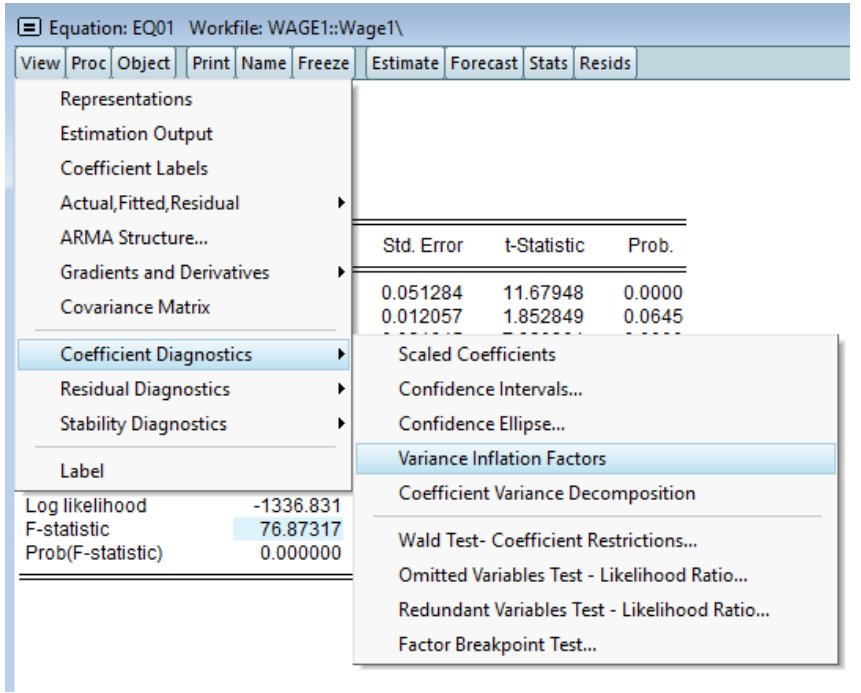
Although the problem of multicollinearity cannot be clearly defined, one thing is clear: everything else being equal, for estimating β_j, it is better to have less correlation between x_j and the other independent variables.

- OLS estimators are still BLUE but would have large covariances.
- Coefficients become “insignificant”.
- Confidence intervals wider (estimates imprecise).
- The OLS results become sensitive to small changes in data.

Multicollinearity in EViews

The variance inflation factor (VIF_j) is the factor by which the variance of β_j is higher because x_j is not uncorrelated with the other explanatory variables.

Setting a cutoff value for VIF above which we conclude multicollinearity is a "problem" is arbitrary and not especially helpful. Sometimes the value 10 is chosen.



Variance Inflation Factors
Date: 12/28/25 Time: 21:27
Sample: 1 526
Included observations: 526

With an intercept

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
EDUC	0.002630	24.06093	1.112771
EXPER	0.000145	3.804971	1.477618
TENURE	0.000468	2.024196	1.349296
C	0.531389	29.37890	NA

Multicollinearity: what to do?

Sometimes there is no choice (data deficiency) – so do nothing.

Drop a variable – but this may lead to specification bias.

Transform variables (take first difference, ratio transformation).

One possible solution is to use **principal components analysis** (PCA). Principal components are used to replace many highly correlated regressors with a small number of their uncorrelated linear combinations that retain most of the variation in the data, thereby reducing multicollinearity at the cost of interpretability.

3. Heteroskedasticity

What is heteroskedasticity?

Homo *(means same or equal)*
Hetero *(means different or unequal)* + skedastic *(means spread or scatter)*

Homoskedasticity means equal spread, …
… and *heteroskedasticity*, on the other hand, means unequal spread.

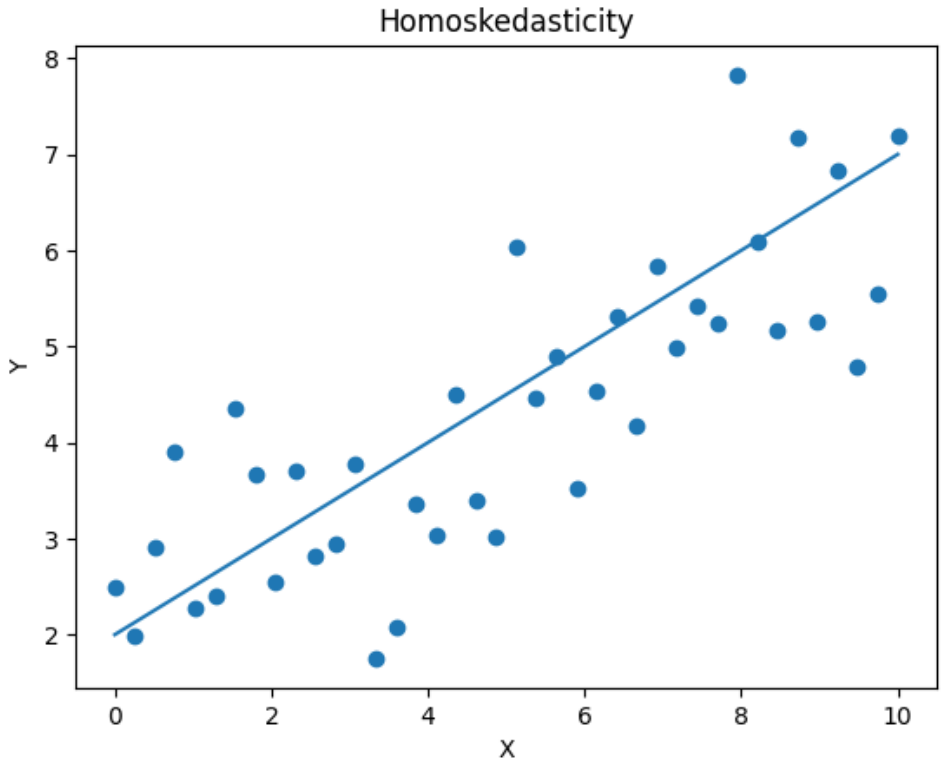
In econometrics the measure we usually use for spread is the variance, and therefore *heteroskedasticity* deals with **unequal variances**.

Assumption 5 of the CLRM assumption was that the variance of the unobserved error, *u*, conditional on the explanatory variables, is constant.

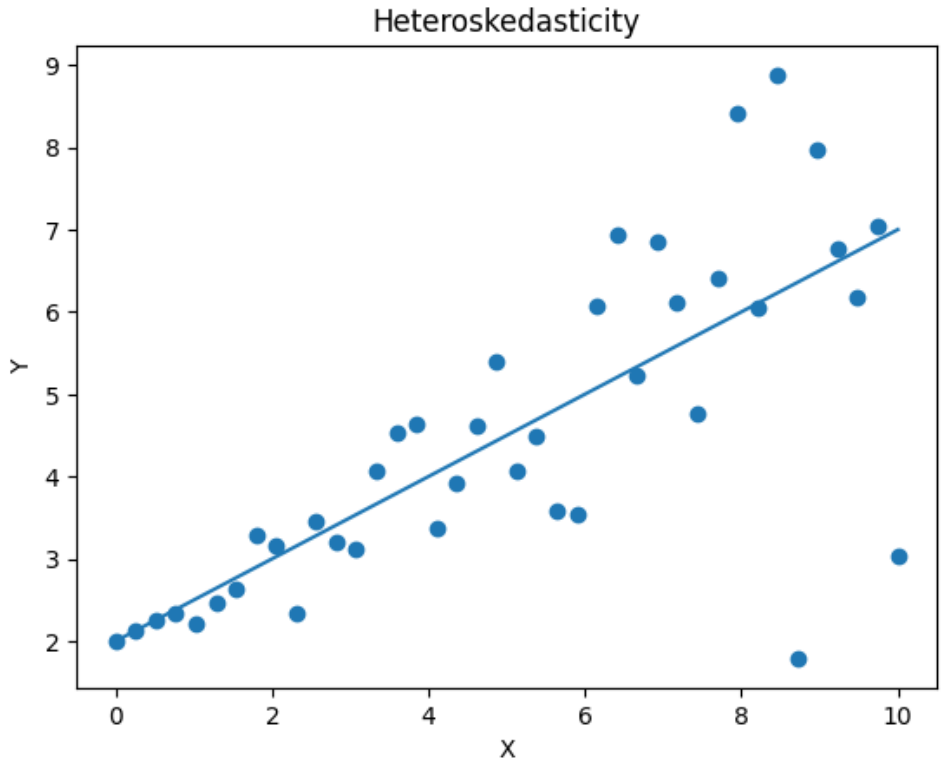
$$\mathrm{Var}(u|x_1, x_2, \dots, x_k) = \sigma^2$$

Example of heteroskedasticity

$$\mathrm{Var}(u_i) = \sigma^2$$



$$\mathrm{Var}(u_i) = \sigma_i^2$$



Consequences of heteroskedasticity for OLS

Heteroskedasticity **does not cause bias or inconsistency** in the OLS estimators of the β_j, whereas something like omitting an important variable would have this effect.

The interpretation of our goodness-of-fit measures, R^2 and $\bar{R}^2$, is also unaffected by the presence of heteroskedasticity.

The estimators of the **variances**, $\mathrm{Var}(\hat{\beta}_j)$, **are biased** without the homoskedasticity assumption.

The usual OLS *t*-statistics do not have *t*-distributions in the presence of heteroskedasticity, and the problem is not resolved by using large sample sizes.

Similarly, *F*-statistics are no longer *F*-distributed.

Detecting heteroskedasticity. The Breusch-Pagan-Godfrey LM test

Let's assume that we have the following model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

The **Breusch-Pagan test** involves the following steps:

1) Run a regression of the model and obtain the residuals $\hat{u}_i$.
2) Run the auxiliary regression of squared residuals $\hat{u}_i^2$:

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \cdots + \delta_k x_k + v$$

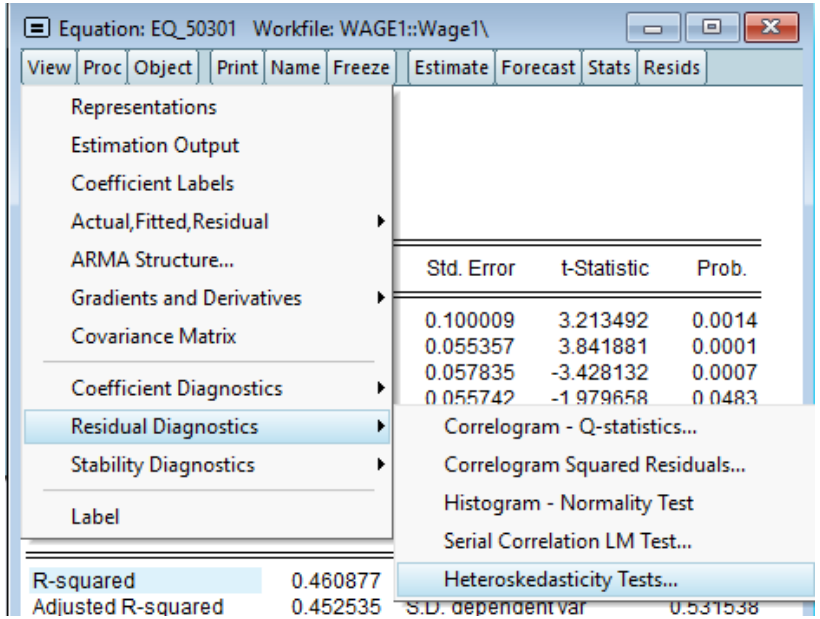
3) Formulate the null and the alternative hypotheses:

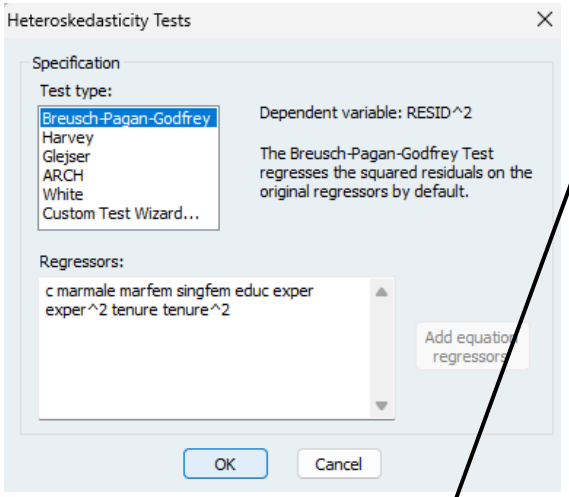
$$\mathrm{H}_0\colon\ \delta_1 = \delta_2 = \cdots = \delta_k = 0 \quad \text{vs} \quad \mathrm{H}_1\colon at\ least\ one\ of \delta_1, \delta_2, \ldots, \delta_k\ is\ different\ from\ 0$$

4) Compute the $LM = nR^2$ statistic, where *n* and R^2 is the number of observations and R-squared in the auxiliary regression from step 2 respectively. The *LM*-statistic follows the χ_k^2 distribution.
5) If the *p*-value of *LM*-statistic is sufficiently small, that is, below the chosen significance level, then we reject the null hypothesis of homoskedasticity.

Detecting heteroskedasticity. The Breusch-Pagan-Godfrey LM test in EViews

Consider the estimated wage model





Heteroskedasticity Test: Breusch-Pagan-Godfrey
Null hypothesis: Homoskedasticity

F-statistic	1.662132	Prob. F(8,517)	0.1050
Obs*R-squared	13.18931	Prob. Chi-Square(8)	0.1055
Scaled explained SS	20.42085	Prob. Chi-Square(8)	0.0089

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 01/06/26 Time: 13:23
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.050286	0.068933	0.729497	0.4660
MARMALE	-0.048700	0.038156	-1.276337	0.2024
MARFEM	-0.051466	0.039864	-1.291035	0.1973
SINGFEM	0.004162	0.038421	0.108314	0.9138
EDUC	0.003849	0.004614	0.834080	0.4046
EXPER	0.010083	0.003614	2.790281	0.0055
EXPER^2	-0.000207	7.61E-05	-2.720426	0.0067
TENURE	0.000476	0.004661	0.102196	0.9186
TENURE^2	8.67E-05	0.000159	0.543936	0.5867

R-squared	0.025075	Mean dependent var	0.152030
Adjusted R-squared	0.009989	S.D. dependent var	0.272446
S.E. of regression	0.271081	Akaike info criterion	0.244167
Sum squared resid	37.99182	Schwarz criterion	0.317148
Log likelihood	-55.21597	Hannan-Quinn criter.	0.272742
F-statistic	1.662132	Durbin-Watson stat	1.947157
Prob(F-statistic)	0.104993		

LM-statistic

p-value = 0.106 > 0.05

We fail to reject H_0: $homoskedasticity$.

Detecting heteroskedasticity. The White test

White (1980) proposed a test for heteroskedasticity that adds the squares and cross products of all the independent variables to the auxiliary equation.

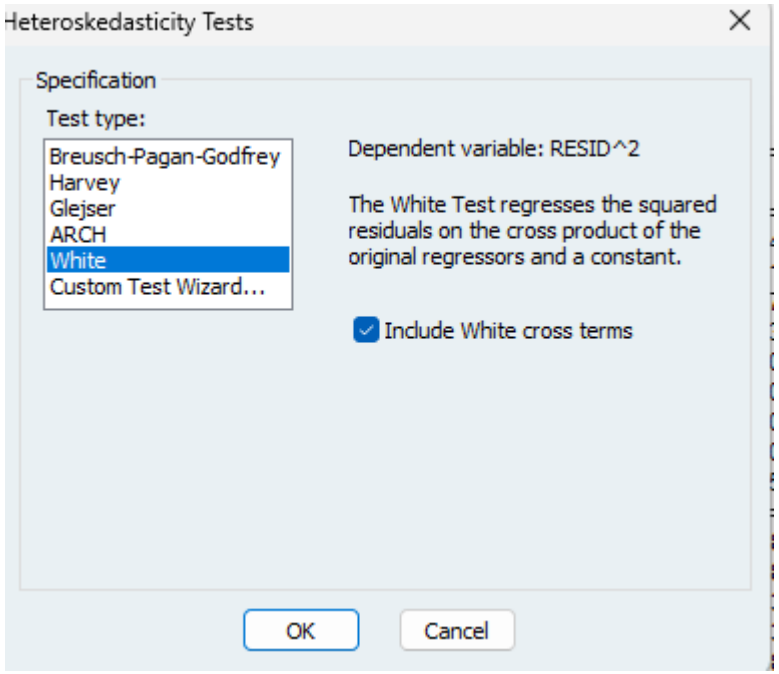
When the model contains $k = 3$ independent variables, the White test is based on an estimation of

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_1^2 + \delta_5 x_2^2 + \delta_6 x_3^2 + \delta_7 x_1 x_2 + \delta_8 x_1 x_3 + \delta_9 x_2 x_3 + error.$$

White test for heteroskedasticity is the *LM*-statistic for testing that all of the δ_j in the equation are zero, except for the intercept.

Detecting heteroskedasticity. The White test in EViews

Consider the estimated wage model



LM-statistic

p-value = 0.129 > 0.05

We fail to reject H_0: *homoskedasticity*.

Heteroskedasticity Test: White
Null hypothesis: Homoskedasticity

F-statistic	1.292045	Prob. F(36,489)	0.1237
Obs*R-squared	45.68727	Prob. Chi-Square(36)	0.1293
Scaled explained SS	70.73706	Prob. Chi-Square(36)	0.0005

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 01/06/26 Time: 13:40
Sample: 1 526
Included observations: 526
Collinear test regressors dropped from specification

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.568176	0.287721	1.974748	0.0489
MARMALE^2	-0.188987	0.252667	-0.747969	0.4548
MARMALE*EDUC	0.008924	0.016474	0.541723	0.5883
MARMALE*EXPER	-0.003912	0.012632	-0.309681	0.7569
MARMALE*EXPER^2	0.000128	0.000263	0.487187	0.6263
MARMALE*TENURE	0.033666	0.020115	1.673663	0.0948
MARMALE*TENURE^2	-0.001687	0.000778	-2.169071	0.0306
MARFEM^2	-0.059867	0.260306	-0.229987	0.8182
MARFEM*EDUC	0.003749	0.017961	0.208715	0.8348
MARFEM*EXPER	-0.010625	0.013494	-0.787421	0.4314
MARFEM*EXPER^2	0.000182	0.000281	0.645847	0.5187
MARFEM*TENURE	0.037780	0.021309	1.772962	0.0769
MARFEM*TENURE^2	-0.001615	0.000839	-1.925119	0.0548
SINGFEM^2	-0.220691	0.209062	-1.055622	0.2917
SINGFEM*EDUC	0.022148	0.016189	1.368103	0.1719
SINGFEM*EXPER	-0.018716	0.011662	-1.604924	0.1092
SINGFEM*EXPER^2	0.000375	0.000263	1.426533	0.1544
SINGFEM*TENURE	0.049676	0.024055	2.065080	0.0394
SINGFEM*TENURE^2	-0.002227	0.001140	-1.953607	0.0513
EDUC^2	0.002499	0.001174	2.127609	0.0339
EDUC*EXPER	-0.000790	0.001557	-0.507228	0.6122
EDUC*EXPER^2	3.22E-05	3.09E-05	1.042122	0.2979
EDUC*TENURE	0.003414	0.002017	1.692804	0.0911
EDUC*TENURE^2	-0.000148	8.32E-05	-1.774823	0.0765
EDUC	-0.071898	0.035584	-2.020495	0.0439
EXPER^2	-0.000440	0.001686	-0.261320	0.7940
EXPER*EXPER^2	-1.37E-05	5.35E-05	-0.255596	0.7984
EXPER*TENURE	-0.002752	0.002058	-1.337303	0.1817
EXPER*TENURE^2	0.000172	0.000115	1.493810	0.1359
EXPER	0.027458	0.027677	0.992090	0.3216
EXPER^2^2	1.52E-07	5.54E-07	0.274678	0.7837
EXPER^2*TENURE	5.75E-05	3.87E-05	1.483716	0.1385
EXPER^2*TENURE^2	-3.26E-06	1.85E-06	-1.765212	0.0782
TENURE^2	0.003160	0.002373	1.331865	0.1835
TENURE*TENURE^2	-9.80E-05	0.000110	-0.891969	0.3728
TENURE	-0.059236	0.034090	-1.737666	0.0829
TENURE^2^2	1.75E-06	1.54E-06	1.129937	0.2591

R-squared	0.086858	Mean dependent var	0.152030
Adjusted R-squared	0.019633	S.D. dependent var	0.272446
S.E. of regression	0.269758	Akaike info criterion	0.285162
Sum squared resid	35.58419	Schwarz criterion	0.585192
Log likelihood	-37.99754	Hannan-Quinn criter.	0.402637
F-statistic	1.292045	Durbin-Watson stat	1.997983
Prob(F-statistic)	0.123674		

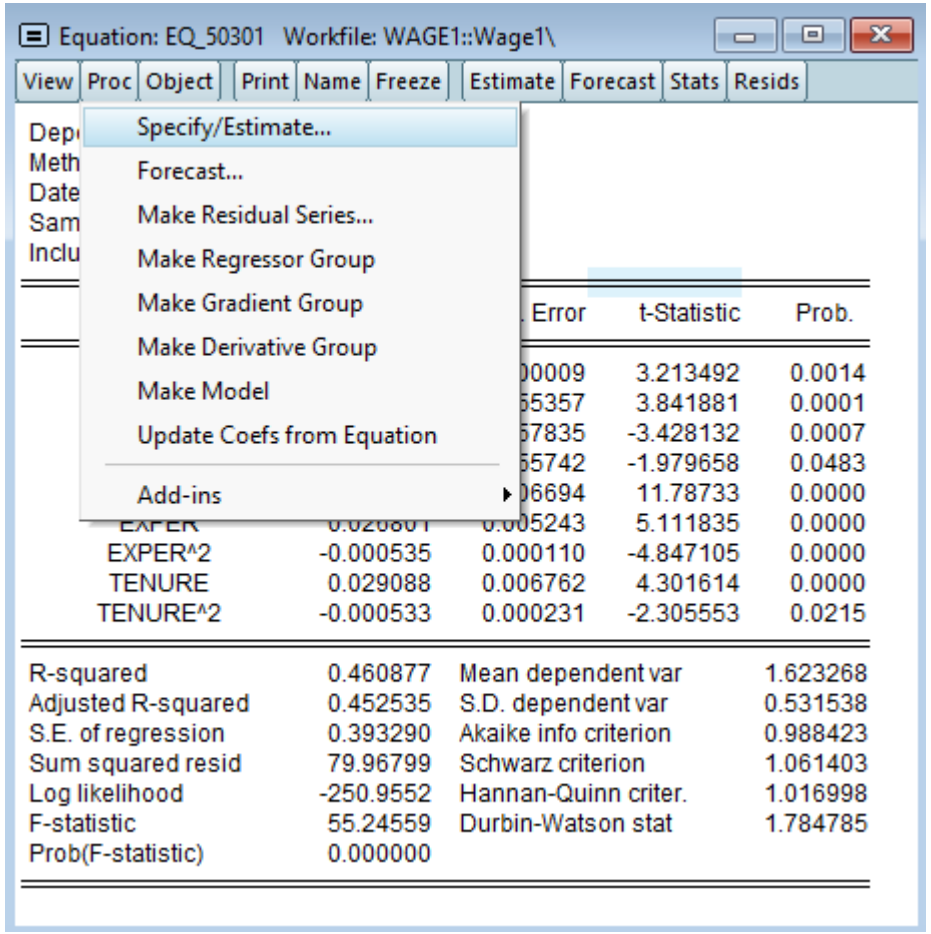
Resolving heteroskedasticity. Heteroskedasticity-robust inference

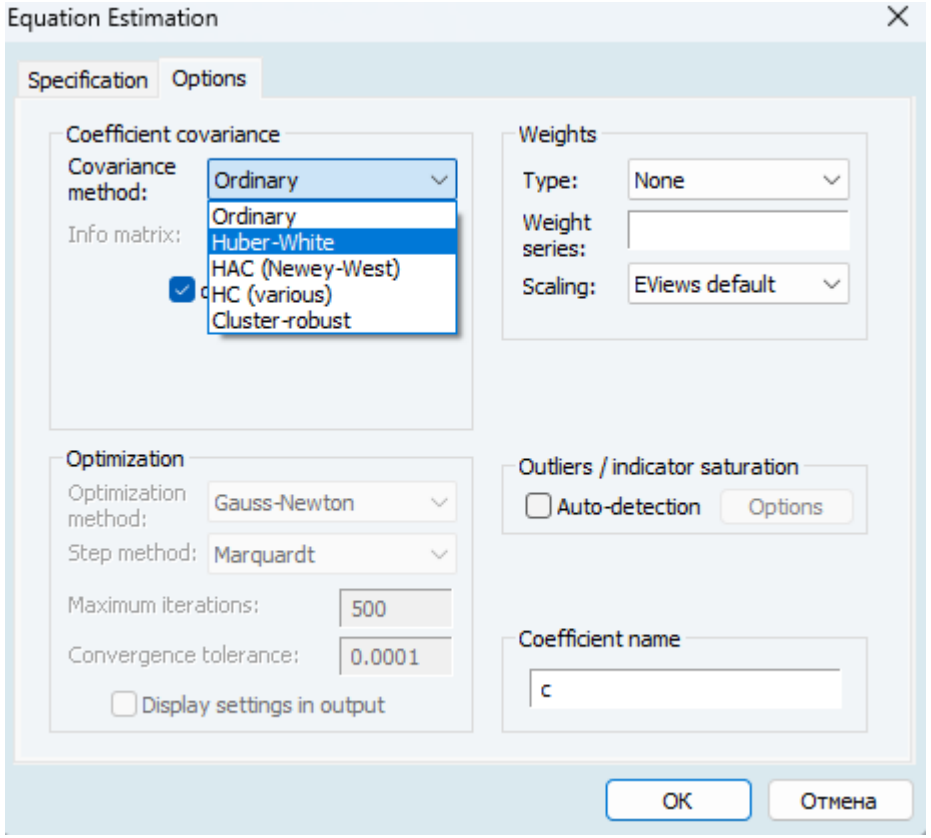
White (1980) proposed a method of obtaining consistent estimators of the variances and covariances of the OLS estimators.

In applied work, these are sometimes called White, Huber, or Eicker standard errors (or some hyphenated combination of these names).

We will just refer to them as **heteroskedasticity-robust standard errors**, or even just **robust standard errors** when the context is clear.

Resolving heteroskedasticity. Robust standard errors in EViews (1)





Resolving heteroskedasticity. Robust standard errors in EViews (2)

Wage model **without** correction

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/06/26 Time: 13:51
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.321378	0.100009	3.213492	0.0014
MARMALE	0.212676	0.055357	3.841881	0.0001
MARFEM	-0.198268	0.057835	-3.428132	0.0007
SINGFEM	-0.110350	0.055742	-1.979658	0.0483
EDUC	0.078910	0.006694	11.78733	0.0000
EXPER	0.026801	0.005243	5.111835	0.0000
EXPER^2	-0.000535	0.000110	-4.847105	0.0000
TENURE	0.029088	0.006762	4.301614	0.0000
TENURE^2	-0.000533	0.000231	-2.305553	0.0215

R-squared	0.460877	Mean dependent var	1.623268
Adjusted R-squared	0.452535	S.D. dependent var	0.531538
S.E. of regression	0.393290	Akaike info criterion	0.988423
Sum squared resid	79.96799	Schwarz criterion	1.061403
Log likelihood	-250.9552	Hannan-Quinn criter.	1.016998
F-statistic	55.24559	Durbin-Watson stat	1.784785
Prob(F-statistic)	0.000000		

Wage model **with** correction

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/06/26 Time: 13:52
Sample: 1 526
Included observations: 526
Huber-White-Hinkley (HC1) heteroskedasticity consistent standard errors and covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.321378	0.109469	2.935791	0.0035
MARMALE	0.212676	0.057142	3.721886	0.0002
MARFEM	-0.198268	0.058770	-3.373619	0.0008
SINGFEM	-0.110350	0.057116	-1.932028	0.0539
EDUC	0.078910	0.007415	10.64246	0.0000
EXPER	0.026801	0.005139	5.215010	0.0000
EXPER^2	-0.000535	0.000106	-5.033361	0.0000
TENURE	0.029088	0.006941	4.190731	0.0000
TENURE^2	-0.000533	0.000244	-2.187835	0.0291

R-squared	0.460877	Mean dependent var	1.623268
Adjusted R-squared	0.452535	S.D. dependent var	0.531538
S.E. of regression	0.393290	Akaike info criterion	0.988423
Sum squared resid	79.96799	Schwarz criterion	1.061403
Log likelihood	-250.9552	Hannan-Quinn criter.	1.016998
F-statistic	55.24559	Durbin-Watson stat	1.784785
Prob(F-statistic)	0.000000	Wald F-statistic	51.69553
Prob(Wald F-statistic)	0.000000		

4. Normality of residuals

Normality of residuals

Recall that one of the assumptions of the CLRM is that the errors are normally distributed with a zero mean and a constant variance. Violation of this assumption leads to the inferential statistics of a regression model (i.e. *t*-stats, *F*-stats, etc.) not being valid.

One way of detecting misspecification problems is through observing the regression residuals.

We can use the **Jarque-Bera statistic** to test the normality of the residuals.

If the *p*-value of the Jarque-Bera statistic is less than 0.05 (for a 5% significance level), then we reject the null hypothesis of normality.

Normality of residuals. Example (1)

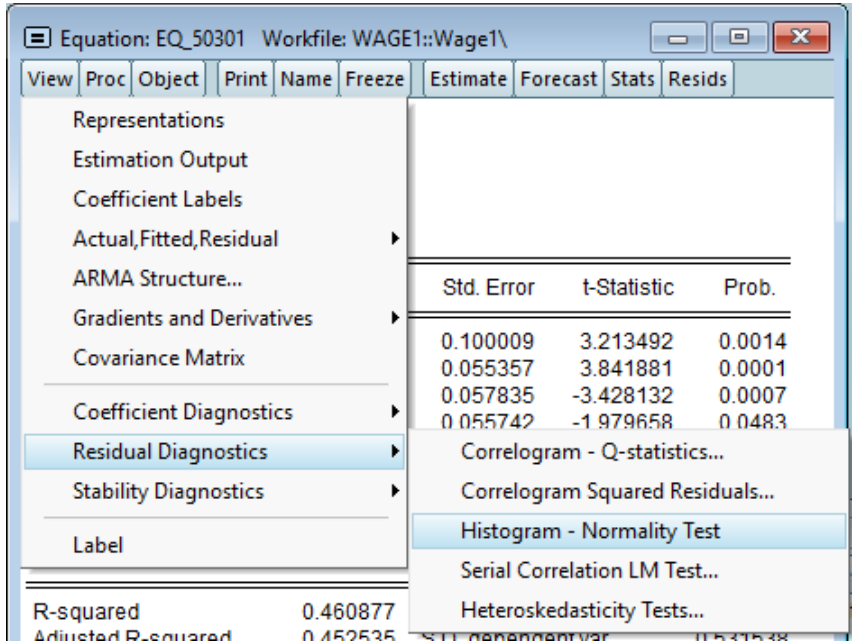
Estimated wage model

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/05/26 Time: 14:45
Sample: 1 526
Included observations: 526

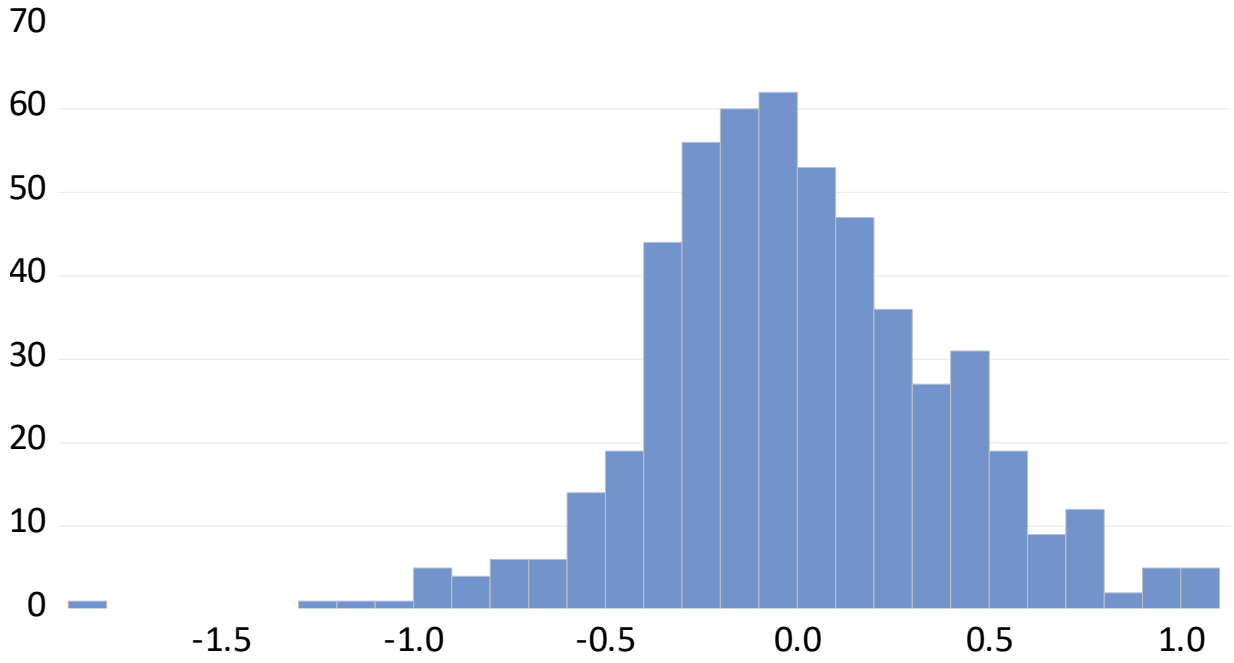
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.321378	0.100009	3.213492	0.0014
MARMALE	0.212676	0.055357	3.841881	0.0001
MARFEM	-0.198268	0.057835	-3.428132	0.0007
SINGFEM	-0.110350	0.055742	-1.979658	0.0483
EDUC	0.078910	0.006694	11.78733	0.0000
EXPER	0.026801	0.005243	5.111835	0.0000
EXPER^2	-0.000535	0.000110	-4.847105	0.0000
TENURE	0.029088	0.006762	4.301614	0.0000
TENURE^2	-0.000533	0.000231	-2.305553	0.0215

R-squared	0.460877	Mean dependent var	1.623268
Adjusted R-squared	0.452535	S.D. dependent var	0.531538
S.E. of regression	0.393290	Akaike info criterion	0.988423
Sum squared resid	79.96799	Schwarz criterion	1.061403
Log likelihood	-250.9552	Hannan-Quinn criter.	1.016998
F-statistic	55.24559	Durbin-Watson stat	1.784785
Prob(F-statistic)	0.000000		

Testing residuals normality



Normality of residuals. Example (2)



Series: Residuals	
Sample 1 526	
Observations 526	
Mean	2.03e-16
Median	-0.026886
Maximum	1.091974
Minimum	-1.896968
Std. Dev.	0.390282
Skewness	-0.062426
Kurtosis	4.205326
Jarque-Bera	32.18240
Probability	0.000000

p-value = 0.00 < 0.05

We reject
H_0: *the residuals are normally distributed.*

Asymptotic normality and large sample inference

However, we can use the central limit theorem to conclude that the OLS estimators satisfy **asymptotic normality**, which means they are approximately normally distributed in large enough sample sizes.

This is useful because the normality assumption MLR.6 has been dropped; the only restriction on the distribution of the error is that it has finite variance (errors are not "wildly explosive"), something we will always assume. We have also assumed zero conditional mean (MLR.4) and homoskedasticity of u (MLR.5).

Unfortunately, there are no general prescriptions on how big the sample size must be before the approximation is good enough. Some econometricians think that $\boldsymbol{n = 30}$ is satisfactory, but this can't be sufficient for all possible distributions of u.

Outliers

An observation is an influential observation if dropping it from the analysis changes the key OLS estimates by a practically “large” amount.

The presence of outliers could lead to non-normality of the residuals.

Outliers can arise when sampling from a small population if one or several members of the population are very different in some relevant aspect from the rest of the population.

R&D intensity and firm size

Suppose that R&D expenditures as a percentage of sales (*rdintens*) are related to *sales* (in millions) and *profits* as a percentage of sales (*profmarg*):

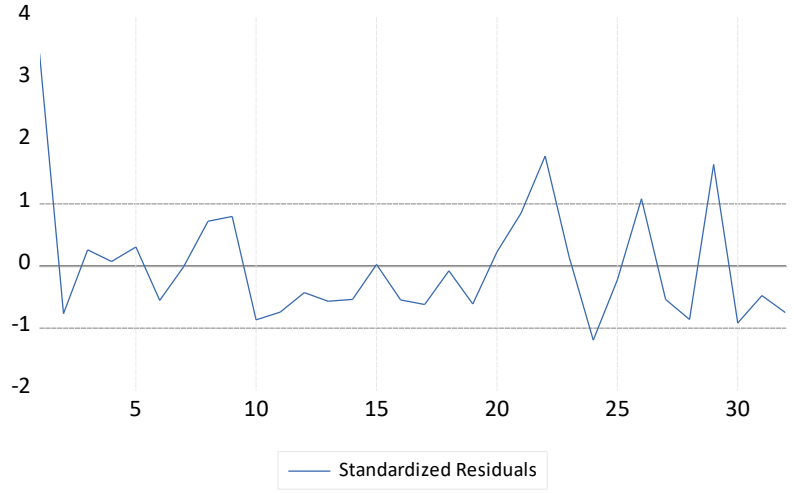
$$rdintens = \beta_0 + \beta_1 sales + \beta_2 profmarg + u$$

The OLS equation using data on 32 chemical companies is

Dependent Variable: RDINTENS
Method: Least Squares
Date: 01/06/26 Time: 15:31
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.625261	0.585533	4.483542	0.0001
SALES	5.34E-05	4.41E-05	1.211091	0.2356
PROFMARG	0.044617	0.046181	0.966134	0.3420

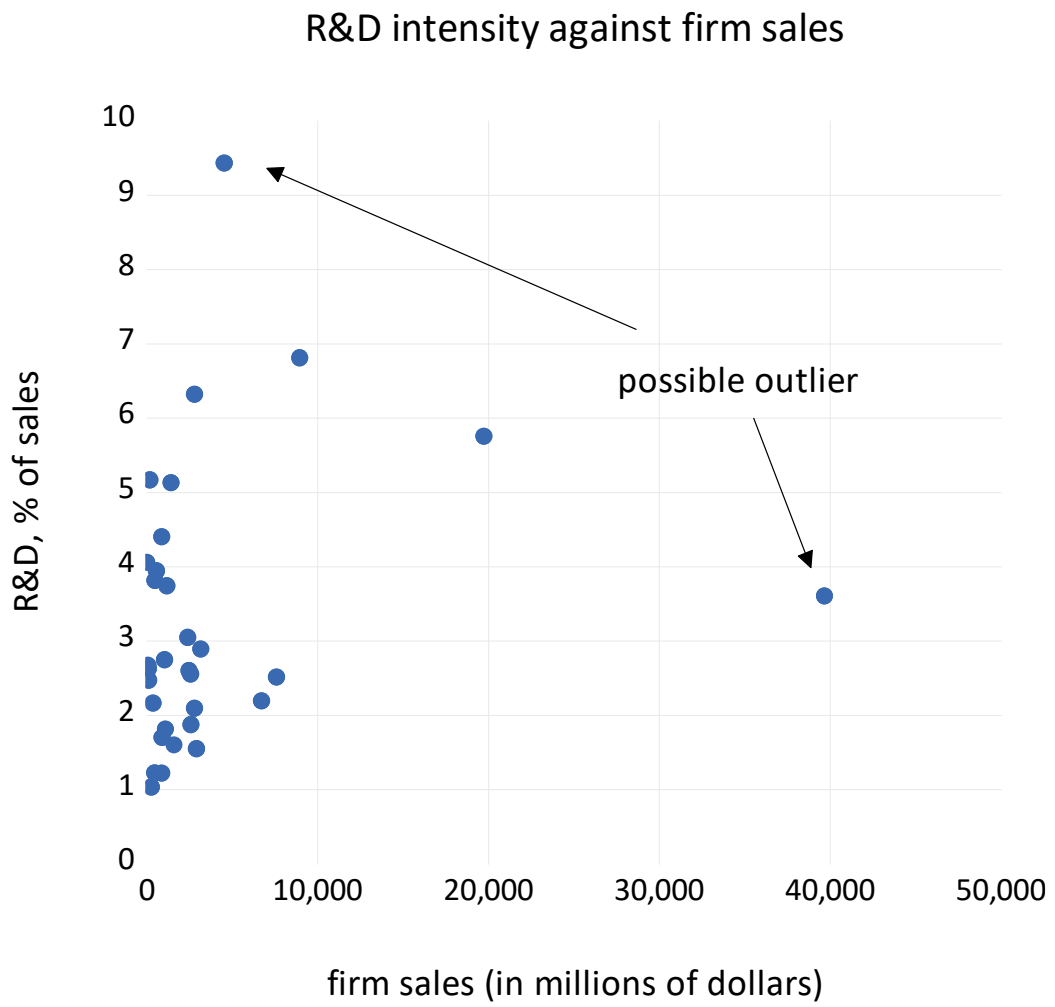
R-squared	0.076118	Mean dependent var	3.266210
Adjusted R-squared	0.012403	S.D. dependent var	1.873701
S.E. of regression	1.862046	Akaike info criterion	4.170289
Sum squared resid	100.5492	Schwarz criterion	4.307701
Log likelihood	-63.72462	Hannan-Quinn criter.	4.215837
F-statistic	1.194653	Durbin-Watson stat	1.694228
Prob(F-statistic)	0.317275		



Series: Residuals
Sample 1 32
Observations 32

Mean	6.11e-16
Median	-0.606771
Maximum	6.370235
Minimum	-2.222148
Std. Dev.	1.800978
Skewness	1.674078
Kurtosis	6.170484
Jarque-Bera	28.34949
Probability	0.000001

R&D intensity and firm size. Potential outliers



Using dummies to eliminate outliers (1)

Create a dummy equals to one for that observation and zero for other observations.

Include the dummy in the regression (using all observation) along with the other explanatory variables.

The baseline OLS equation

Dependent Variable: RDINTENS
Method: Least Squares
Date: 01/06/26 Time: 15:31
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.625261	0.585533	4.483542	0.0001
SALES	5.34E-05	4.41E-05	1.211091	0.2356
PROFMARG	0.044617	0.046181	0.966134	0.3420

R-squared	0.076118	Mean dependent var	3.266210
Adjusted R-squared	0.012403	S.D. dependent var	1.873701
S.E. of regression	1.862046	Akaike info criterion	4.170289
Sum squared resid	100.5492	Schwarz criterion	4.307701
Log likelihood	-63.72462	Hannan-Quinn criter.	4.215837
F-statistic	1.194653	Durbin-Watson stat	1.694228
Prob(F-statistic)	0.317275		

The OLS equation with dummy for 10th firm

Dependent Variable: RDINTENS
Method: Least Squares
Date: 01/06/26 Time: 15:41
Sample: 1 32
Included observations: 32

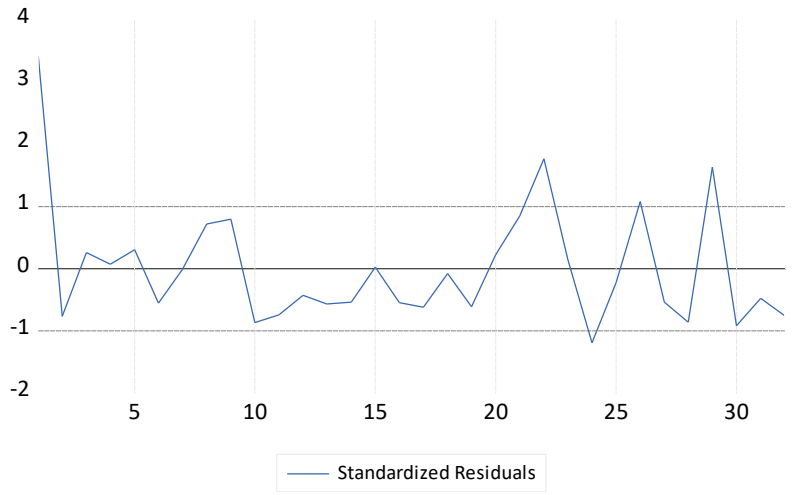
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.296851	0.591804	3.881097	0.0006
SALES	0.000186	8.42E-05	2.204394	0.0359
PROFMARG	0.047841	0.044483	1.075489	0.2913
DUM_10	-6.571716	3.614727	-1.818039	0.0798

R-squared	0.173664	Mean dependent var	3.266210
Adjusted R-squared	0.085128	S.D. dependent var	1.873701
S.E. of regression	1.792176	Akaike info criterion	4.121207
Sum squared resid	89.93305	Schwarz criterion	4.304424
Log likelihood	-61.93931	Hannan-Quinn criter.	4.181938
F-statistic	1.961501	Durbin-Watson stat	1.721434
Prob(F-statistic)	0.142667		

Using dummies to eliminate outliers (2)

Sometimes, outliers are defined by the size of the residual in an OLS regression, where all of the observations are used.

Original model



The OLS equation with dummy for 1st firm

Dependent Variable: RDINTENS
Method: Least Squares
Date: 01/06/26 Time: 15:41
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.193379	0.461463	4.753103	0.0001
SALES	5.06E-05	3.40E-05	1.487455	0.1481
PROFMARG	0.068295	0.035996	1.897283	0.0681
DUM_1	6.718103	1.474875	4.555033	0.0001

R-squared	0.469342	Mean dependent var	3.266210
Adjusted R-squared	0.412486	S.D. dependent var	1.873701
S.E. of regression	1.436182	Akaike info criterion	3.678322
Sum squared resid	57.75333	Schwarz criterion	3.861539
Log likelihood	-54.85315	Hannan-Quinn criter.	3.739053
F-statistic	8.254899	Durbin-Watson stat	1.909225
Prob(F-statistic)	0.000432		

The OLS equation with dummy for 10th firm

Dependent Variable: RDINTENS
Method: Least Squares
Date: 01/06/26 Time: 15:41
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.296851	0.591804	3.881097	0.0006
SALES	0.000186	8.42E-05	2.204394	0.0359
PROFMARG	0.047841	0.044483	1.075489	0.2913
DUM_10	-6.571716	3.614727	-1.818039	0.0798

R-squared	0.173664	Mean dependent var	3.266210
Adjusted R-squared	0.085128	S.D. dependent var	1.873701
S.E. of regression	1.792176	Akaike info criterion	4.121207
Sum squared resid	89.93305	Schwarz criterion	4.304424
Log likelihood	-61.93931	Hannan-Quinn criter.	4.181938
F-statistic	1.961501	Durbin-Watson stat	1.721434
Prob(F-statistic)	0.142667		

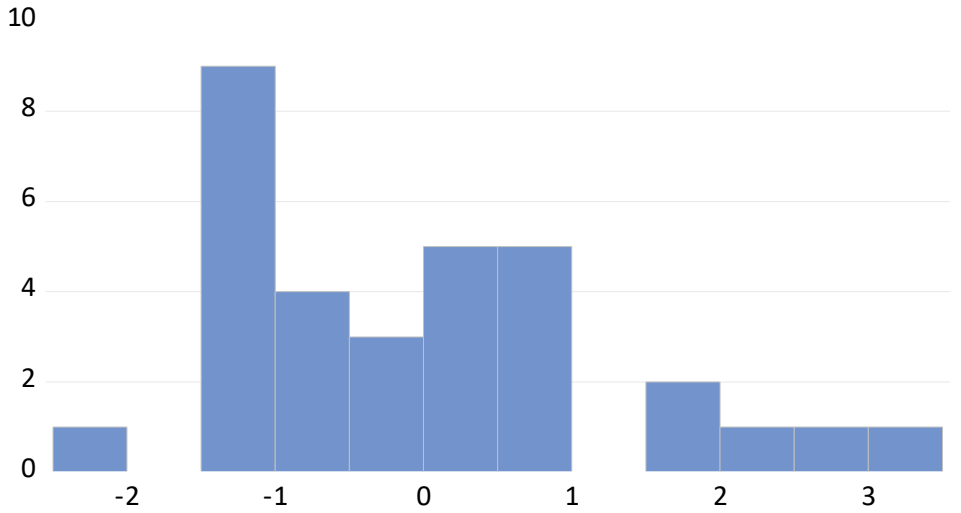
Using dummies to eliminate outliers (3)

The OLS equation with dummies for 1st and 10th firms

Dependent Variable: RDINTENS
Method: Least Squares
Date: 01/06/26 Time: 15:45
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.938958	0.458789	4.226255	0.0002
SALES	0.000160	6.46E-05	2.472019	0.0200
PROFMARG	0.070066	0.034326	2.041200	0.0511
DUM_1	6.466894	1.411823	4.580526	0.0001
DUM_10	-5.414306	2.772871	-1.952600	0.0613

R-squared	0.535004	Mean dependent var	3.266210
Adjusted R-squared	0.466116	S.D. dependent var	1.873701
S.E. of regression	1.369065	Akaike info criterion	3.608734
Sum squared resid	50.60714	Schwarz criterion	3.837755
Log likelihood	-52.73974	Hannan-Quinn criter.	3.684648
F-statistic	7.766249	Durbin-Watson stat	1.880845
Prob(F-statistic)	0.000266		



Series: Residuals
Sample 1 32
Observations 32

Mean	2.36e-16
Median	-0.189095
Maximum	3.143880
Minimum	-2.066708
Std. Dev.	1.277689
Skewness	0.836170
Kurtosis	3.075357
Jarque-Bera	3.736529
Probability	0.154391

Functional forms and outliers (1)

For most economic variables, the *logarithmic transformation* significantly narrows the range of the data and also yields functional forms – such as constant elasticity models – that can explain a broader range of data.

R&D spending (no logarithms)

Dependent Variable: RD
Method: Least Squares
Date: 01/07/26 Time: 09:36
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-9.012926	33.02535	-0.272909	0.7869
SALES	0.040630	0.002486	16.34422	0.0000
PROFMARG	0.857292	2.604685	0.329135	0.7444

R-squared	0.902092	Mean dependent var	153.6813
Adjusted R-squared	0.895340	S.D. dependent var	324.6358
S.E. of regression	105.0235	Akaike info criterion	12.23531
Sum squared resid	319868.2	Schwarz criterion	12.37272
Log likelihood	-192.7649	Hannan-Quinn criter.	12.28085
F-statistic	133.5988	Durbin-Watson stat	2.084978
Prob(F-statistic)	0.000000		

R&D spending (logarithms)

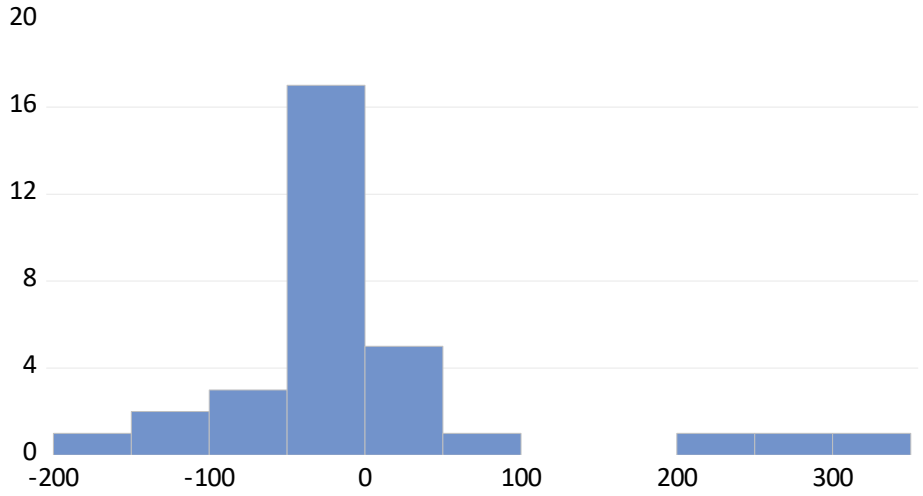
Dependent Variable: LOG(RD)
Method: Least Squares
Date: 01/07/26 Time: 09:36
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-4.378273	0.468018	-9.354916	0.0000
LOG(SALES)	1.084220	0.060195	18.01179	0.0000
PROFMARG	0.021656	0.012783	1.694150	0.1010

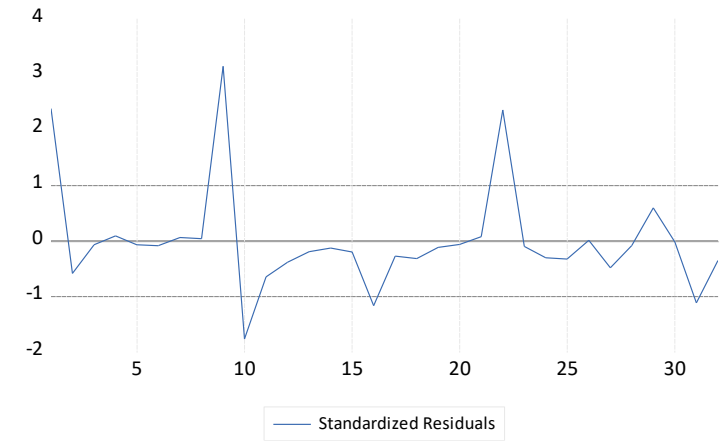
R-squared	0.917955	Mean dependent var	3.602825
Adjusted R-squared	0.912296	S.D. dependent var	1.734352
S.E. of regression	0.513625	Akaike info criterion	1.594413
Sum squared resid	7.650510	Schwarz criterion	1.731826
Log likelihood	-22.51062	Hannan-Quinn criter.	1.639962
F-statistic	162.2314	Durbin-Watson stat	1.858993
Prob(F-statistic)	0.000000		

Functional forms and outliers (2)

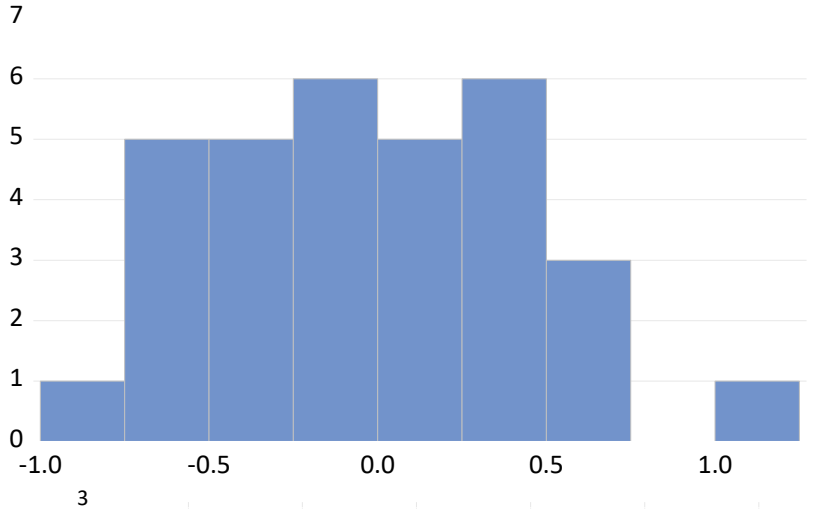
R&D spending (no logarithms)



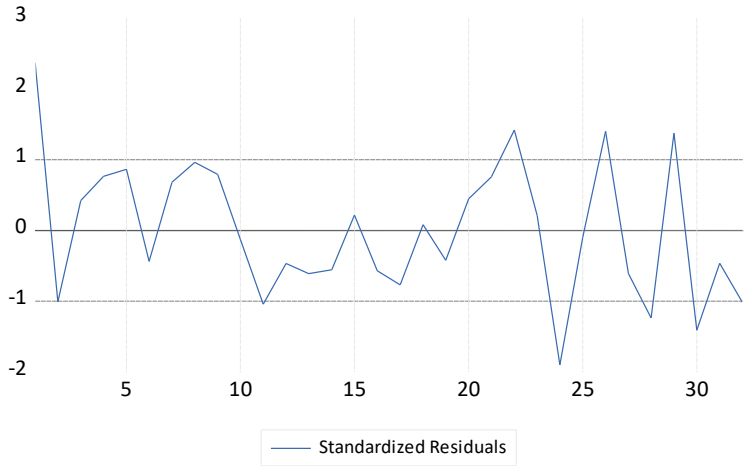
Series: Residuals	
Sample 1 32	
Observations 32	
Mean	-1.07e-14
Median	-11.08015
Maximum	330.5219
Minimum	-185.3353
Std. Dev.	101.5792
Skewness	1.769109
Kurtosis	6.659584
Jarque-Bera	34.54873
Probability	0.000000



R&D spending (logarithms)



Series: Residuals	
Sample 1 32	
Observations 32	
Mean	-1.49e-15
Median	-0.058281
Maximum	1.217832
Minimum	-0.976811
Std. Dev.	0.496780
Skewness	0.308018
Kurtosis	2.648609
Jarque-Bera	0.670634
Probability	0.715111



Violation of the CLRM assumptions. Summary

Assumption	Violation may imply	Possible solutions
MLR.1 Linearity of the model	Wrong regressors Nonlinearity Changing parameters	Transform variables (logs, etc.) Use nonlinear methods of estimation
MLR.2 Random sampling	Endogenous sample selection	Design your sampling so selection doesn't depend on *y*
MLR.3 No perfect collinearity *(more realistic – strong imperfect correlation)*	Multicollinearity	Transform variables Drop a variable *(may lead to bias)* PCA
MLR.4 Zero conditional mean	Functional form misspecification Omitted variables Reverse causality Measurement error in regressors	Transform variables, use quadratics Use a proxy for the omitted variable Instrumental variables Difference-in-differences
MLR.5 Homoskedasticity	Heteroskedasticity	Use robust standard errors Generalized least squares
MLR.6 Normality of disturbance	Outliers Wrong functional form	Transform variables (logs, etc.) Eliminate outliers